

Logical Networks and SDN in Azure Local



Agenda

- **Objective:** Review logical networks and core SDN functionality on Azure Local
- **Scope:** Logical vs virtual networks, SDN Enabled by Azure Arc vs SDN Managed by On-Prem Tools, implementing SDN Enabled by Azure Arc.



Logical and Virtual Networks in Azure Local

- Logical networks (including SDN-based ones) = Azure Local VM connectivity only
- SDN virtual networks = advanced overlay networking, Hyper-V VMs only
- Logical networks map directly to **physical VLANs**
- SDN virtual networks use **overlays (VNI)** and **network virtualization**

Logical Networks vs. Virtual Networks

- **Logical Networks**
 - Map directly to VLANs
 - Provide subnets + IP pools
 - Used for the most common VM workloads
 - Available without SDN and with SDN enabled by Azure Arc
 - Support Azure Local VMs only
- **Virtual Networks**
 - Use overlay encapsulation
 - Provide isolation, NAT, overlapping IPs
 - Intended for complex SDN scenarios
 - Available with SDN managed by on-premises tools
 - Support Hyper-V VMs only

Logical Networks: Core Definition

- Logical networks represent physical networks for VM provisioning
- Define:
 - Address prefixes
 - Subnets
 - VLAN IDs
 - IP pools
 - Gateways & DNS
- All Azure Local VMs connect using logical networks

SDN Virtual Networks: Core Definition

- Provide overlay-based network virtualization
- Decouple VM networks from physical topology
- Enable:
 - Overlapping IPs
 - NAT/LB services
 - Tenant isolation
 - Advanced routing

Logical Network Types

- **Infrastructure Logical Network**
 - Created automatically
 - Hosts Azure Arc resource bridge VM - no workload Azure Local VMs
 - Includes **management IP range**
 - **Not modifiable** (can be deleted only when decommissioning instance)
- **Workload Logical Networks**
 - Created by admins
 - Host workload Azure Local VMs
 - Support **DNS updates**, IP pools, VLANs

Requirements & Constraints of Logical Networks

- Azure Local instance deployed & registered
- Requires an external VM switch
- Deployed via Azure CLI, Portal, or Terraform
- Support IPv4 only

Creating Logical Networks

- **Components of a logical network:**
 - VLAN ID
 - Address prefixes
 - Gateway
 - DNS servers
 - vSwitch name
- **To implement a static network:**
 - Set IP allocation = Static
- **To implement DHCP:**
 - Set IP allocation = Dynamic
 - Set up a DHCP server and DHCP scopes
 - Define DHCP options (e.g. DNS, gateway, etc.)

Immutable Settings After Creation

- Plan logical networks carefully (most parameters are immutable)
- Once created, no updates are supported for:
 - VLAN ID
 - IP pools
 - Address space
 - Gateway
 - vSwitch name

DNS Server Updates

- Supported for workload logical networks
- Restrictions:
 - Not supported directly for **DHCP logical networks** (must be handled via DHCP)
 - Not supported for networks hosting **AKS** deployments
 - Affect only **VMs** provisioned after the update (in-guest update is required for existing VMs)

Azure Local SDN Basics

- **SDN = centralized control of:**
 - Switching
 - Routing
 - Firewalling
 - Load balancing
- **Can be implemented on Azure Local as:**
 - SDN Enabled by Azure Arc
 - SDN Managed by On-Prem Tools
- **Network Controller provides:**
 - Programmable control plane
 - Southbound API to network devices
 - Northbound API (REST) to Azure/Tools
 - Plumbs policies into VFP-enabled vSwitch

SDN Enabled by Azure Arc

- Network Controller runs as **Failover Cluster service**
- **Managed via:**
 - Azure portal
 - Azure CLI
 - ARM templates
- **Supports:**
 - Logical networks
 - VM NICs
 - NSGs

SDN Managed by On-Prem Tools

- Network Controller runs as **Failover Cluster clustered VM(s)**
- **Managed via:**
 - Windows Admin Center
 - SDN Express scripts
 - SCVMM
 - PowerShell
- **Supports:**
 - Virtual networks
 - SLB, Gateways
 - NSGs for unmanaged Hyper-V VMs

SDN Management Comparison

Capability	SDN Enabled by Arc	SDN Managed by On-Premises Tools
Supported SDN Resources	Logical networks VM NICs NSGs	Logical networks VM NICs NSGs Virtual networks SLBs VPN Gateways
Supported VMs	Azure Local VMs	Hyper-V VMs SCVMM VMs
Management Tools	Azure portal Azure CLI ARM templates	SDN Express scripts Windows Admin Center PowerShell SCVMM

Network Security Groups (NSGs)

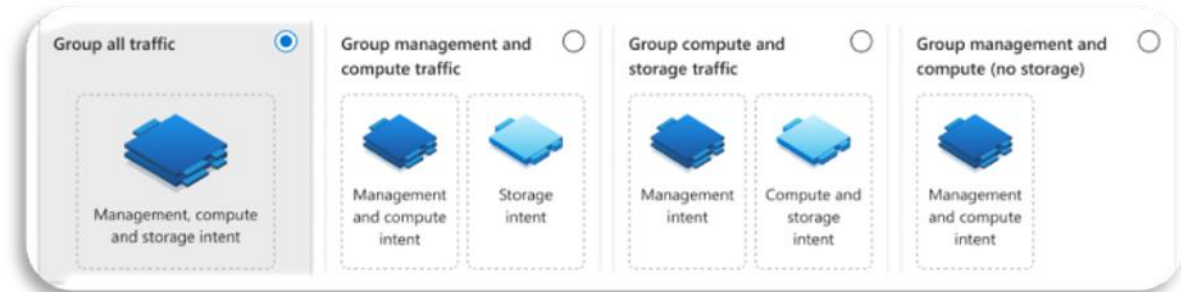
- **NSGs support:**
 - Allow/Deny rules
 - Filtering based on IP, port, protocol, direction
 - Association with logical networks or VM NICs
- **Each rule defines:**
 - Priority (100–4096)
 - Source/Destination IPs
 - Ports
 - Protocol (TCP/UDP/*)
 - Access (Allow/Deny)
 - Direction

Unsupported Scenarios (SDN Enabled by Azure Arc)

- Virtual networks
- SLB
- VPN Gateways
- NSGs on AKS networks
- Multicast

Supported Intent Patterns (SDN Enabled by Azure Arc)

- Group all traffic on one intent
- Mgmt+Compute, separate Storage
- Custom (disaggregated) configuration



Unsupported Intent Patterns (SDN Enabled by Azure Arc)

- More than 3 intents
- Combined **Compute+Storage** intent
- Standalone compute on single node
- Three intents on **2-node switchless** clusters

Enabling SDN: Prerequisites

- Azure Local 2506+
- OS 26100.xxxx+
- SDNPrefix meeting naming rules:
 - (≤ 8 chars, allowed characters, hyphen rules)
- **DNS preparation:**
 - A record for the Network Controller REST URL (<SDNPrefix>-NC) points to the reserved IP (the 5th IP in the reserved range from deployment).
- **Azure Stack HCI Admin RBAC role**

Enabling SDN via PowerShell

- Run:

```
Add-EceFeature -Name NC -SDNPrefix <prefix>
```

- Outcome:
 - Network Controller deployed as **Failover Cluster Service**
 - Existing logical networks hydrated automatically

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